

Appendix

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1. Introduction

This document serves as an appendix to the paper titled "Evaluating human-in-the-loop machine-learning (HITL-ML) querying strategies for error correction"¹. This appendix includes the results from the model-feature hyperparameter grid search in Section 2, the classification results in Section 3, the regression results in Section 4, and a summary of the results in Section 5.

2. Grid search results

Hyperparameters are selected following a ten-fold cross-validation grid search for each model-feature combination. Grid searches are conducted on all available data, withholding data used for the holdout sets. The results for the decision tree, k-NN, and MLP algorithms are shown in Tables 1, 2, and 3, respectively.

Table 1: Decision Tree Grid Search Results

Dat aset	Feature	Problem Type	Criterion	Max Dep th	Min Samp les Leaf	Split ter	Metri c	Best Scor e
HVB	Material Weight	Regressi on	squared _error	10	1	best	R2	1.00
	Material Part Number	Regressi on	squared _error	10	1	best	R2	0.99
	Minimu m Capacity	Regressi on	squared _error	15	1	rand om	R2	1.00
	Capacity Through put	Regressi on	absolute _error	20	1	best	R2	1.00
	Voltage	Regressi on	friedma n_mse	7	32	best	R2	0.75
	Model Code	Classific ation	gini	10	1	rand om	Accur acy	1.00
Red win	quality	Classific ation	gini	15	1	best	Accur acy	0.62

¹ Sheni, D.N., Basson, A.H., Grobler, J.: Evaluating human-in-the-loop machine-learning (HITL-ML) querying strategies for error correction (2025). Submitted to Elsevier Knowledge-based Systems 2025.

Dat aset	Feature	Problem Type	Criterion	Max Dep th	Min Samp les Leaf	Split ter	Metri c	Best Scor e
e- qual ity								
Abal one	Rings	Regressi on	squared _error	10	32	best	R2	0.60

For the k-NN algorithm (Table 2), the cosine distance metric is selected for most features and shows significantly better training performance than other distance metrics in the conducted experiments.

Table 2: k-NN Grid Search Results

Datas et	Feature	Problem Type	Metric (Distan ce)	Neighbo urs	Weigh ts	Metric	Bes t Sco re
HVB	Material Weight	Regressio n	cosine	3	distan ce	R2	1.0 0
	Material Part Number	Regressio n	cosine	3	distan ce	R2	0.9 2
	Minimu m Capacit y	Regressio n	cosine	3	distan ce	R2	0.9 9
	Capacit y Through put	Regressio n	cosine	3	distan ce	R2	0.7 6
	Voltage	Regressio n	cosine	11	unifor m	R2	0.7 1
	Model Code	Classifica tion	manhat tan	3	distan ce	Accur acy	0.7 0
Red wine- qualit y	quality	Classifica tion	cosine	11	distan ce	Accur acy	0.6 6
Abalo ne	Rings	Regressio n	cosine	15	distan ce	R2	0.5 9

The results for the MLP algorithm are shown in Table 3. The adaptive moment estimation (Adam) solver was selected for all MLP-based active learners, as the datasets are large [61]. The maximum iterations were tuned to 2000 to balance model fit and computational effort. L2 regularisation was tested separately, and an ‘alpha’ value of 0.0001 was selected.

Table 3: MLP Grid Search Results

Datas et	Feature	Problem Type	Activati on	Hidd en Layer s	Learni ng Rate	Metric	Best Scor e
HVB	Material Weight	Regression	logistic	(130, 50)	0.00	R2	0.91
	Material Part Number	Regression	relu	(70, 10)	0.00	R2	0.37
	Minimum Capacity	Regression	logistic	(100, 70)	0.00	R2	0.94
	Capacity Through put	Regression	identity	(150, 10)	0.01	R2	0.21
	Voltage	Regression	identity	(90, 50)	0.01	R2	0.32
	Model Code	Classification	logistic	(70, 10)	0.00	Accura cy	0.82
Red wine- qualit y	quality	Classification	logistic	(50, 130)	0.00	Accura cy	0.57
Abalo ne	Rings	Regression	logistic	(80, 30)	0.00	R2	0.27

3. Additional classification results

Figure 1 shows the sample performance of the three model types on the “Quality” feature from the red wine dataset. The decision tree learners generally outperform other model types, exhibiting robust performance. While model types such as the k-NN improve as more queries are retrieved, the performance does not reach that of the decision tree models that are trained on the initial training data. Although models such as k-NN improved as more queries were added, their performance did not reach that of decision trees trained on the initial dataset. Model performance can be improved by removing noise from the training sets, such as noise created by querying similar data.

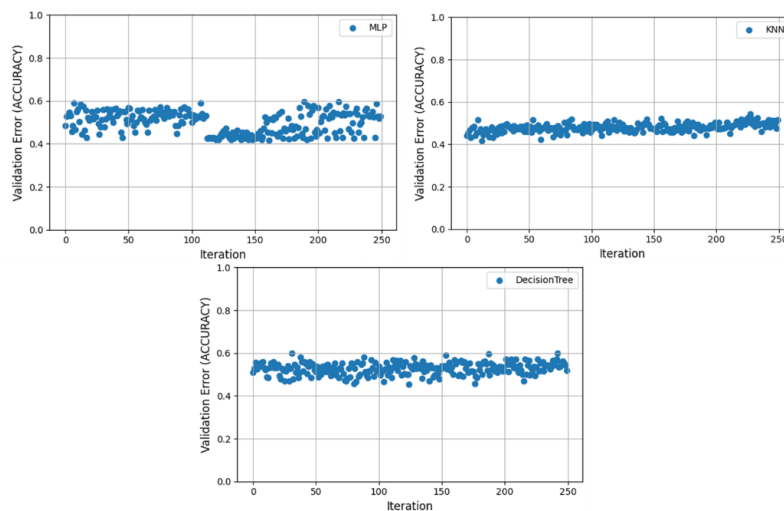


Figure 1: Sample distributions of model performance for the accuracy on the “Quality” feature, grouped by model type, with the MLP algorithm exhibiting catastrophic forgetting (top-left), the KNN algorithm showing robust performance and minor improvement in accuracy as the active learning process continues (top-right), and the decision tree algorithm exhibiting robust performance similar to a random distribution (bottom).

3.1 Red wine – quality

Table 4 shows the mean accuracy results for the active learners on the red wine dataset.

Table 4: Accuracy results on the red wine quality dataset over 30 repetitions

Learner (Query Strategy)	Accuracy (%)	Standard Deviation (%)
Decision tree (QBC)	53.42	0.31

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Learner (Query Strategy)	Accuracy (%)	Standard Deviation (%)
Decision tree (Least confidence)	• 53.39	• 0.25
Decision tree (Margin sampling)	• 53.33	• 0.28
Decision tree (Random)	• 53.32	• 0.30
Decision tree (Entropy sampling)	• 53.13	• 0.52
MLP (random)	• 49.38	• 0.37
MLP (Least Confidence)	• 49.21	• 0.41
MLP (Margin Sampling)	• 49.19	• 0.42
MLP (Entropy Sampling)	• 49.15	• 0.38
MLP (QBC)	• 48.88	• 1.29
KNN (QBC)	• 48.56	• 0.72
KNN (Entropy Sampling)	• 48.51	• 0.69
KNN (Least Confidence)	• 48.50	• 0.89
KNN (random)	• 48.49	• 0.99
KNN (Margin Sampling)	• 48.37	• 0.79

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The Decision tree (QBC and least confidence) learners are tied for first place based on the statistical significance. The entropy sampling yields the lowest performance across model types on this feature. Table 5 shows the Mann-Whitney U test results.

Table 5: Summarised Mann-Whitney U test results for model-strategy pairs

Learner (Query Strategy)	Total Wins	Total Losses	Total Draws	Performance Score
Decision tree (QBC)	11	0	3	0.89
Decision tree (Least confidence)	11	0	3	0.89
Decision tree (Margin sampling)	10	0	4	0.86
Decision tree (Random)	10	0	4	0.86
Decision tree (Entropy sampling)	10	2	2	0.79
MLP (Random)	6	5	3	0.54
MLP (QBC)	5	5	4	0.50
MLP (Least confidence)	5	5	4	0.50
MLP (Margin sampling)	5	5	4	0.50
MLP (Entropy sampling)	5	6	3	0.46
KNN (QBC)	0	10	4	0.14
KNN (Least confidence)	0	10	4	0.14

KNN (Random)	0	10	4	0.14
KNN (Margin sampling)	0	10	4	0.14
KNN (Entropy sampling)	0	10	4	0.14

Figure 2 shows the accuracy, as a percentage, across all active learners and per model type for the “Quality” feature. The QBC strategy consistently has a better performance, with a better distribution and higher maximum values. The entropy sampling strategy shows the opposite, with relatively longer tails and less robust performance.

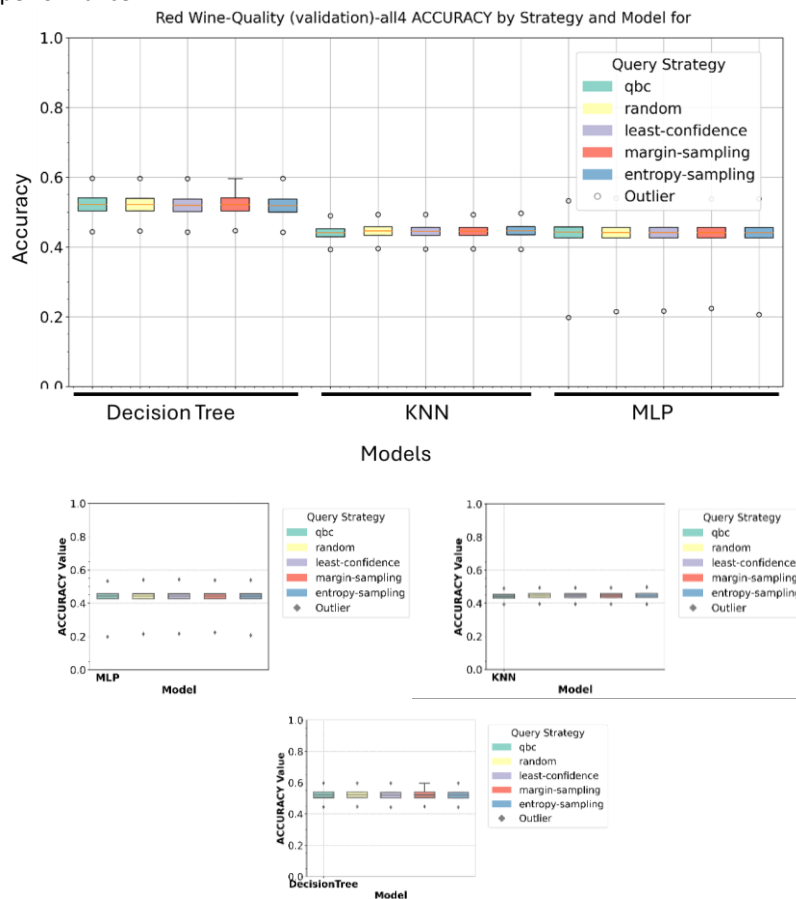


Figure 2: Accuracy results from all active learners (top) and per model type (bottom) on the Quality feature from the Red wine dataset

3.2 Model Code

Table 6 shows the accuracy and standard deviation for each active learner.

Table 6: Mean accuracy and standard deviation over 30 repetitions on the 'Model Code' feature of the HVB dataset

Learner (Strategy)	Accuracy (%)	Standard Deviation (%)
Decision Tree (Random)	98.94	0.07
Decision Tree (Least Confidence)	98.84	0.29
Decision Tree (QBC)	98.83	0.34
Decision Tree (Margin Sampling)	98.82	0.35
Decision Tree (Entropy Sampling)	98.82	0.27
KNN (Entropy Sampling)	62.88	0.28
KNN (QBC)	62.87	0.25
KNN (Least Confidence)	62.87	0.30
KNN (Margin Sampling)	62.84	0.25
KNN (Random)	62.83	0.19
MLP (QBC)	58.43	1.82
MLP (Entropy Sampling)	58.38	1.60
MLP (Margin Sampling)	57.91	1.45
MLP (Random)	57.69	1.75
MLP (Least Confidence)	57.59	1.63

The decision tree (Random) learner dominates in this dataset, being the only statistically significant result of its model type, with the QBC learners of the k-NN and MLP have slightly better performance in their specific model types, but

results are not found to be statistically significant. Table 7 shows the Mann-Whitney U test results for the “Model Code” feature.

Table 7: Mann-Whitney U-test results for the accuracy metric on the Model Code feature from the HVB dataset

Learner (Strategy)		Total Wins		Total Losses		Total Draws		Performance Score
Decision tree (Random)	•	11	•	0	•	3	•	0.89
Decision tree (QBC)	•	10	•	0	•	4	•	0.86
Decision tree (Margin sampling)	•	10	•	0	•	4	•	0.86
Decision tree (Least confidence)	•	10	•	0	•	4	•	0.86
Decision tree (Entropy sampling)	•	10	•	1	•	3	•	0.82
KNN (QBC)	•	5	•	5	•	4	•	0.50
KNN (Margin sampling)	•	5	•	5	•	4	•	0.50
KNN (Least confidence)	•	5	•	5	•	4	•	0.50
KNN (Random)	•	5	•	5	•	4	•	0.50
KNN (Entropy sampling)	•	5	•	5	•	4	•	0.50
MLP (QBC)	•	0	•	10	•	4	•	0.14
MLP (Margin sampling)	•	0	•	10	•	4	•	0.14
MLP (Least confidence)	•	0	•	10	•	4	•	0.14
MLP (Random)	•	0	•	10	•	4	•	0.14
MLP (Entropy sampling)	•	0	•	10	•	4	•	0.14

Figure 3 shows the accuracy, as a percentage, across all active learners and per model type for the “Model Code” feature. The random sampling seems to show more robust performance in this dataset. The QBC performance seems to be less robust in this feature, indicating a relatively even distribution of class instances, attributed to the larger dataset size, as compared to the red wine dataset.

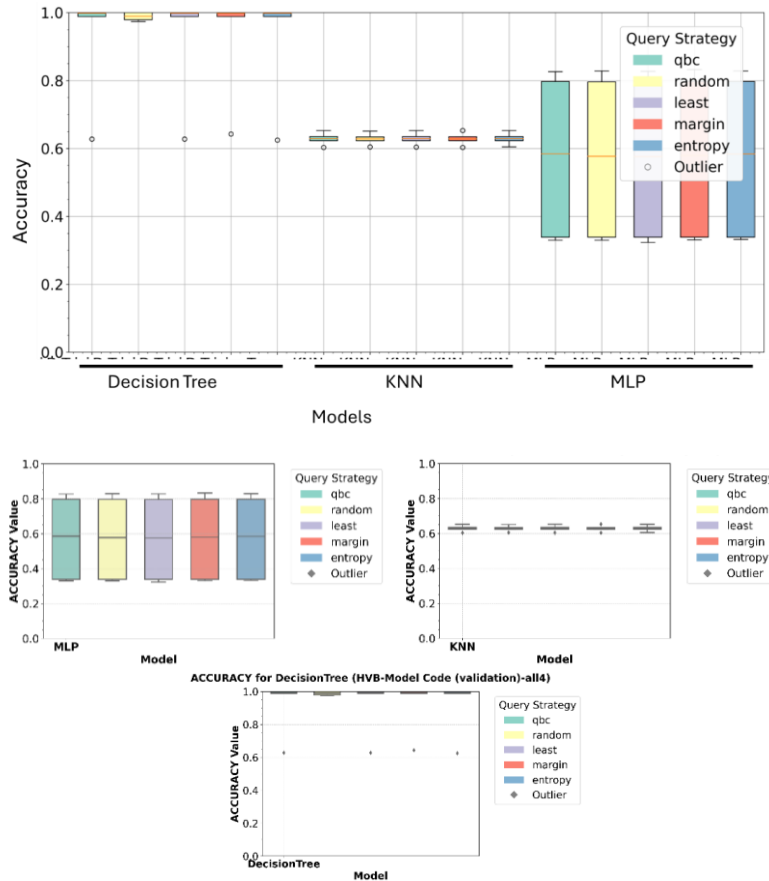


Figure 3: Accuracy results from all active learners (top) and per model type (bottom) on the Model Code feature from the HVB dataset

4. Additional regression results

RMSE, MAE, and R^2 distributions and distributions for R^2 for each model type are included under the relevant feature subheading.

On the HVB dataset, QBC outperformed random sampling across most models and features. Decision trees achieved the highest R^2 scores (≥ 0.99) and the lowest NRMSEs on “Minimum Capacity” and “Capacity Throughput” (Tables 5 and 6), with QBC again slightly outperforming random sampling. KNN also performed well, while MLPs had the lowest model performance. The rankings are consistent across all metrics for each feature. Each features learner rankings,

4.1 Abalone – Rings

Table 8 shows the results for the “rings” feature on the abalone dataset.

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Table 8: Mean results on the Rings feature from the abalone dataset over 30 repetitions

Learner (Strategy)		NRMSE		Standard Deviation		R2		Standard Deviation
Decision Tree (QBC)	•	0.08	•	0.00	•	0.53	•	0.01
Decision Tree (random)	•	0.08	•	0.00	•	0.52	•	0.01
KNN (QBC)	•	0.08	•	0.00	•	0.52	•	0.00
KNN (random)	•	0.08	•	0.00	•	0.51	•	0.00
MLP (QBC)	•	0.15	•	0.00	•	-0.91	•	0.16
MLP (random)	•	0.15	•	0.01	•	-0.96	•	0.24

Decision tree (QBC) achieved the highest performance score, winning all of its pairwise comparisons, while MLPs underperformed across both strategies. Notable is the KNN (QBC) learner tied in second place with the decision tree (Random) learner. Table 9 shows the Mann-Whitney U results.

Table 9: Mann-Whitney U-test results rankings on the Rings feature from the abalone dataset

Model (Strategy)	Total Wins	Total Losses	Total Draws	Performance Score
Decision tree (QBC)	5	0	0	1.00
KNN (QBC)	3	1	1	0.70
Decision tree (random)	3	1	1	0.70
KNN (Random)	2	3	0	0.40
MLP (QBC)	0	4	1	0.10
MLP (Random)	0	4	1	0.10

Figures of RMSE, MAE, and R^2 are shown in Figures 4, 5, and 6, respectively.

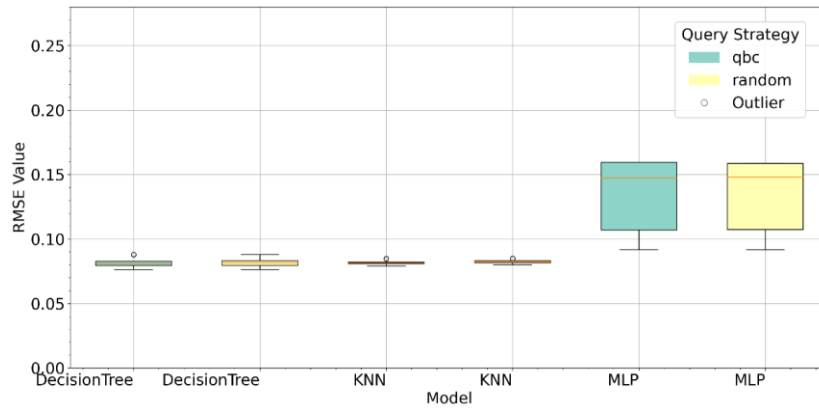


Figure 4: RMSE results for all active learners on the “Rings” feature.

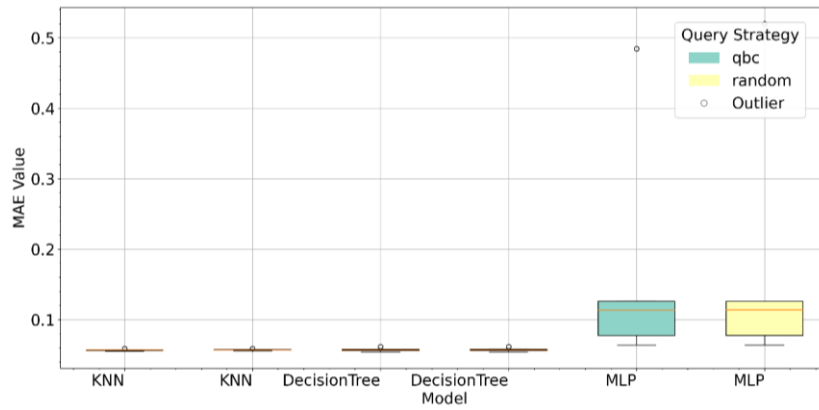


Figure 5: MAE results for all active learners on the “Rings” feature.

Figure 6 shows the R^2 scores for the active learners on the abalone dataset.

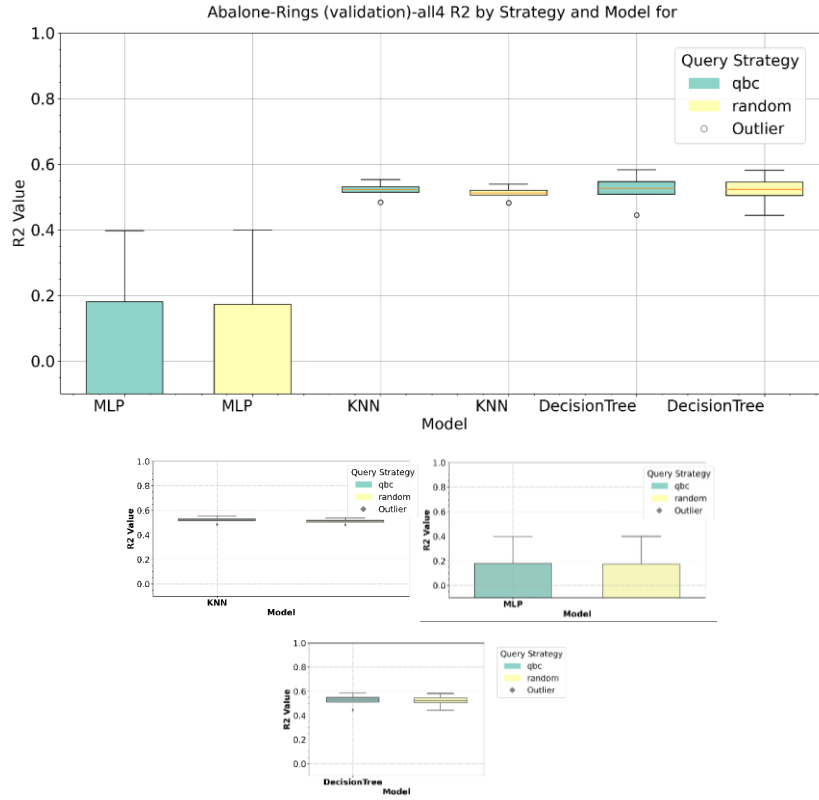


Figure 6: R2 results from all active learners (top) and per model type (bottom) on the Rings feature from the abalone dataset

4.2 Capacity Throughput

Table 10 shows the results of the active learners on the “Capacity Throughput” feature of the HVB dataset

Table 10: Mean results on Capacity Throughput

Learner (Strategy)		NRMSE		Standard deviation		R2		Standard deviation
Decision Tree (QBC)	•	0.01	•	0.00	•	0.99	•	0.00
Decision Tree (random)	•	0.01	•	0.00	•	0.99	•	0.00
KNN (QBC)	•	0.05	•	0.00	•	0.71	•	0.00
KNN (Random)	•	0.05	•	0.00	•	0.70	•	0.00

Learner (Strategy)		NRMSE		Standard deviation		R2		Standard deviation
MLP (QBC)	•	0.10	•	0.00	•	-	•	0.05
						0.31		
MLP (Random)	•	0.10	•	0.00	•	-	•	0.08
						0.35		

Decision tree (QBC) achieved the top performance score, outperforming all other learners in its pairwise comparisons. The random strategy version of the decision tree also performed strongly, ranking second. MLPs, particularly under random sampling, consistently underperformed. KNN (QBC) achieved moderate performance but was outperformed by both decision tree variants. The Mann-Whitney U results are shown in Table 11.

Table 11: Mann-Whitney U-test results rankings on the Capacity Throughput feature from the HVB dataset

Model (Strategy)		Total Wins		Total Losses		Total Draws		Performance Score
Decision tree (QBC)	•	5	•	0	•	0	•	1.00
Decision tree (Random)	•	4	•	1	•	0	•	0.80
KNN (QBC)	•	3	•	2	•	0	•	0.60
KNN (Random)	•	2	•	3	•	0	•	0.40
MLP (QBC)	•	1	•	4	•	0	•	0.20
MLP (Random)	•	0	•	5	•	0	•	0.00

Figures of RMSE, MAE, and R^2 are shown in Figure 7, Figure 8, and Figure 9, respectively. There is a clear difference between the QBC variants and the random sampling variants, with statistical significance confirmed for each model type.

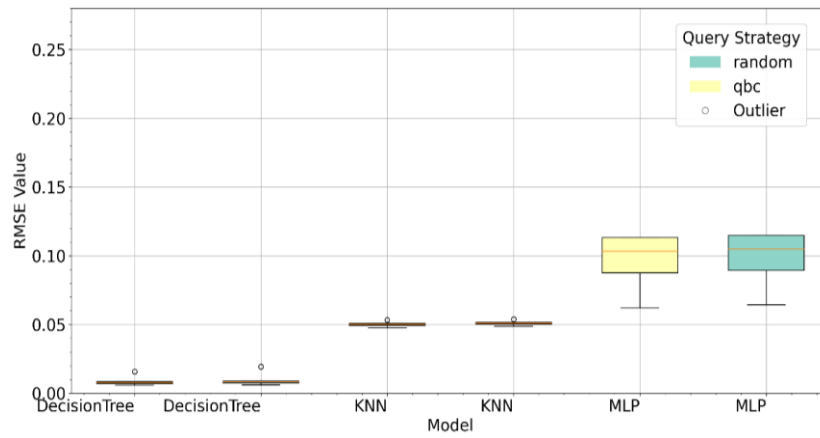


Figure 7: RMSE results for all active learners on the “Capacity Throughput” feature.

The MAE and R^2 results for all active learners are shown in Figure 8 and Figure 9, respectively.

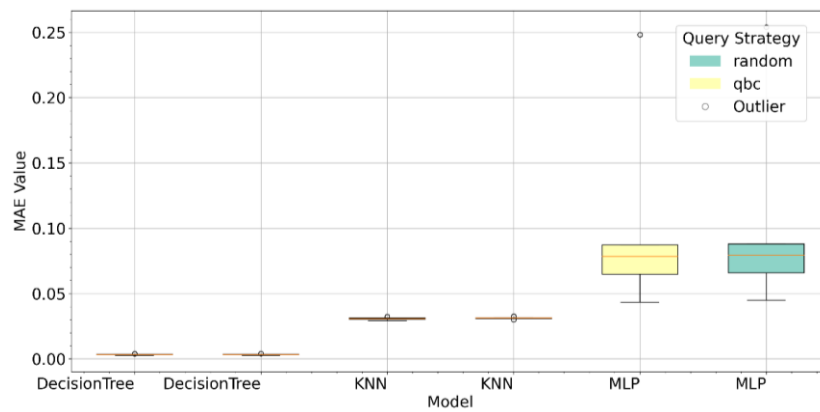


Figure 8: MAE results for all active learners on the “Capacity Throughput” feature

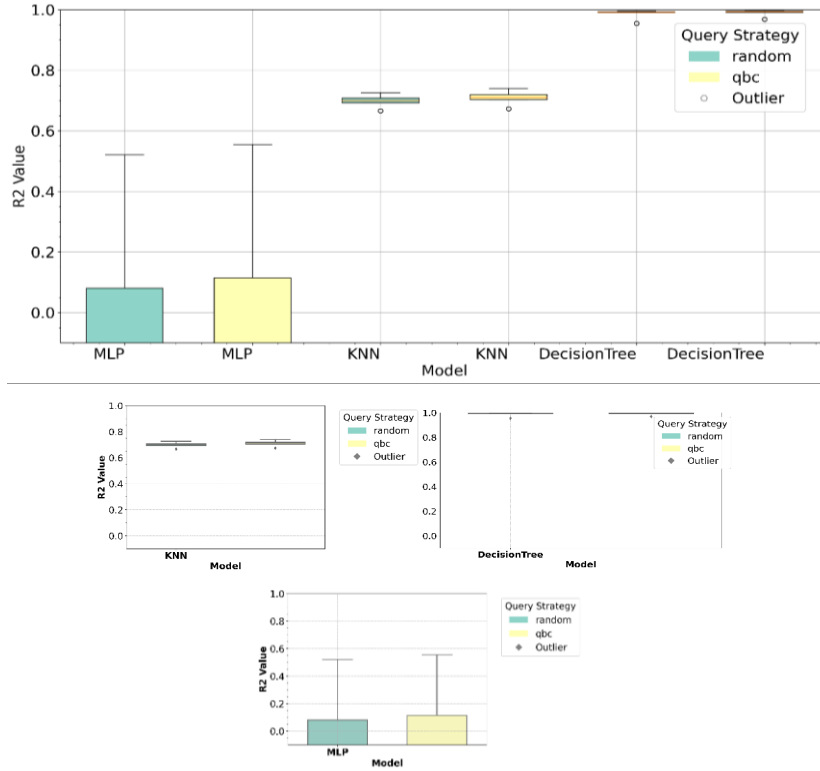


Figure 9: R² results from all active learners (top) and per model type (bottom) on the Capacity Throughput feature from the HVB dataset.

4.3 Minimum Capacity

Table 12 contains the results for the active learners on the “Minimum Capacity” feature of the HVB dataset.

Table 12: Mean results on minimum capacity

Learner (Strategy)	NRMSE		Standard deviation		R2		Standard Deviation	
Decision Tree (QBC)	•	0.00	•	0.00	•	1.00	•	0.00
Decision Tree (random)	•	0.00	•	0.00	•	1.00	•	0.00
KNN (QBC)	•	0.02	•	0.00	•	0.99	•	0.00
KNN (Random)	•	0.02	•	0.00	•	0.99	•	0.00
MLP (Random)	•	0.11	•	0.01	•	0.76	•	0.03
MLP (QBC)	•	0.11	•	0.01	•	0.75	•	0.04

Decision tree (Random) and decision tree (QBC) learners tied for the highest performance score, each winning four pairwise comparisons and drawing once. KNN (QBC) outperformed its random counterpart, while both MLP variants underperformed, with identical scores. These results demonstrate consistency in decision tree robustness across querying strategies. Mann-Whitney U tests are shown in Table 13.

Table 13: Mann-Whitney U-test ranking results on the Minimum Capacity feature from the HVB dataset

Learner (Strategy)		Total Wins		Total Losses		Total Draws		Performance Score
Decision tree (Random)	•	4.00	•	0.00	•	1.00	•	0.90
Decision tree (QBC)	•	4.00	•	0.00	•	1.00	•	0.90
KNN (QBC)	•	3.00	•	2.00	•	0.00	•	0.60
KNN (Random)	•	2.00	•	3.00	•	0.00	•	0.40
MLP (Random)	•	0.00	•	4.00	•	1.00	•	0.10
MLP (QBC)	•	0.00	•	4.00	•	1.00	•	0.10

The distributions of MAE, RMSE, and R^2 are shown in Figure 10, Figure 11, and

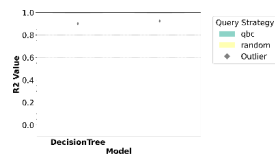


Figure 12, respectively. Only the querying k-NN (QBC) learner outperforms its random sampling variant. The MLP (Random) learn yields lower error and a better R^2 score, but this is not confirmed to be statistically significant.

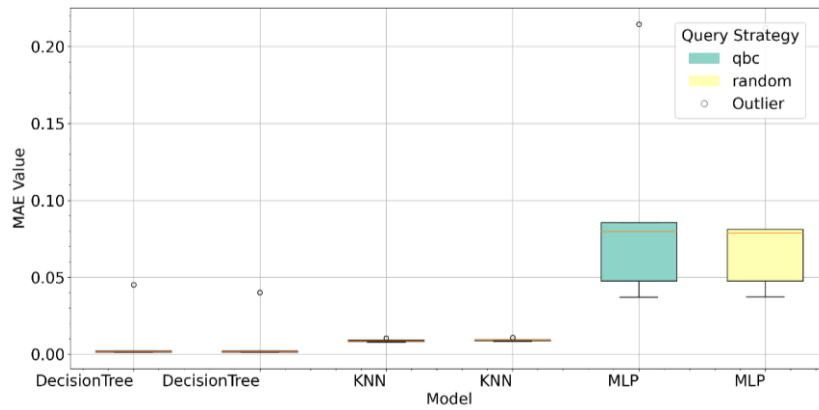


Figure 10: MAE results for all active learners on the “Minimum Capacity” feature.

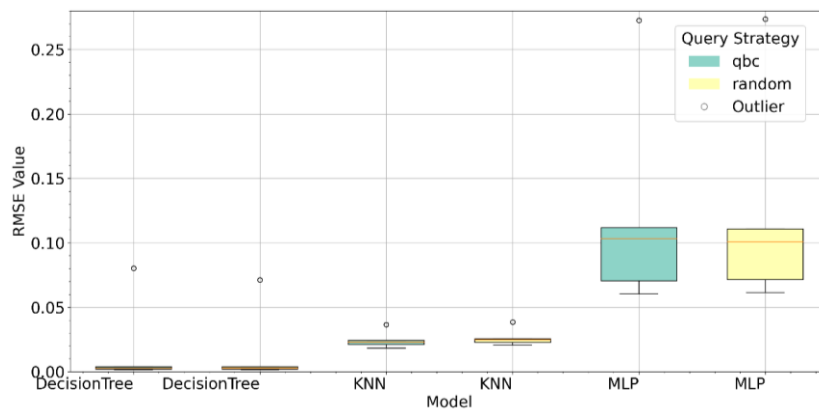


Figure 11: RMSE results for all active learners on the “Minimum Capacity” feature.

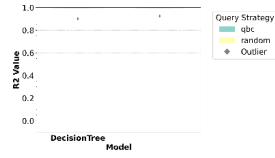


Figure 12 presents the R^2 results. On the Minimum Capacity feature.

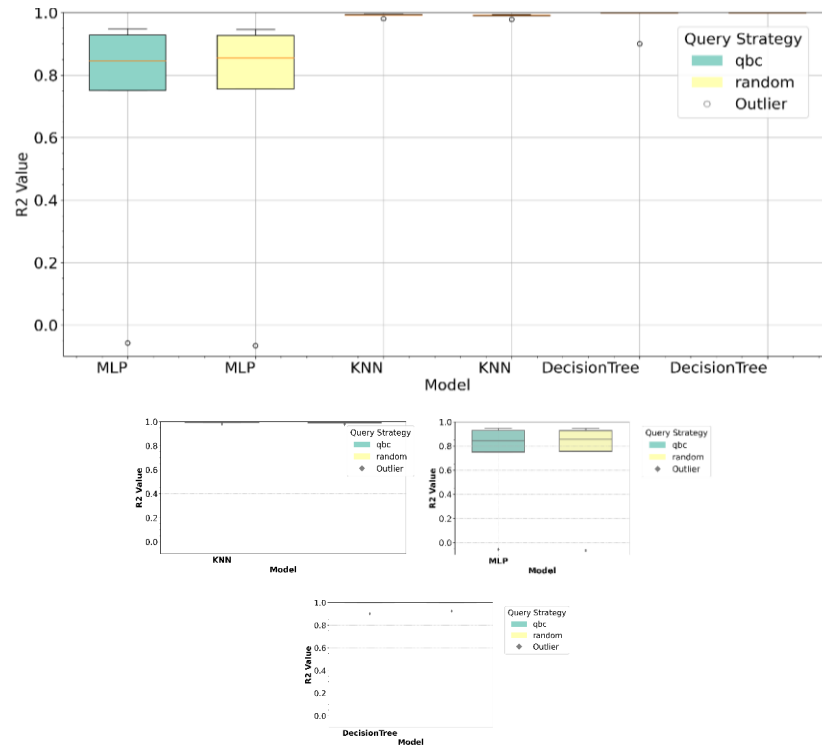


Figure 12: R^2 results from all active learners (top) and per model type (bottom) on the Minimum Capacity feature from the HVB dataset.

4.4 Voltage

The results of the active learners on the “Voltage” feature from the HVB dataset are shown in Table 14.

Table 14: Mean results on voltage

Learner (Query Strategy)	NRMSE	Standard deviation	R^2	Standard deviation
Decision Tree (Random)	0.10	0.00	0.71	0.00

Decision Tree (QBC)	0.10	0.00	0.71	0.00
KNN (QBC)	0.11	0.00	0.68	0.00
KNN (Random)	0.11	0.00	0.67	0.00
MLP (QBC)	0.18	0.00	0.06	0.02
MLP (Random)	0.18	0.00	0.05	0.02

Decision tree learners, using both QBC and random sampling, tied for the top performance score, each winning four comparisons and drawing one. KNN (QBC) again outperformed KNN (Random), consistent with previous features. The MLP (QBC) learner outperforms its random counterpart. These findings further highlight the reliability of decision trees and the added value of QBC for k-NN and MLP. Mann-Whitney U results are shown in Table 15.

Table 15: Mann-Whitney U-test results rankings on the Voltage feature from the HVB dataset

Learner (Strategy)	Total Wins		Total Losses		Total Draws		Performance Score
Decision tree (QBC)	•	4	•	0	•	1	• 0.90
Decision tree (Random)	•	4	•	0	•	1	• 0.90
KNN (QBC)	•	3	•	2	•	0	• 0.60
KNN (Random)	•	2	•	3	•	0	• 0.40
MLP (QBC)	•	1	•	4	•	0	• 0.20
MLP (Random)	•	0	•	5	•	0	• 0.00

Figure 13, Figure 14, and Figure 15 show the RMSE, MAE, and R^2 distributions, respectively.

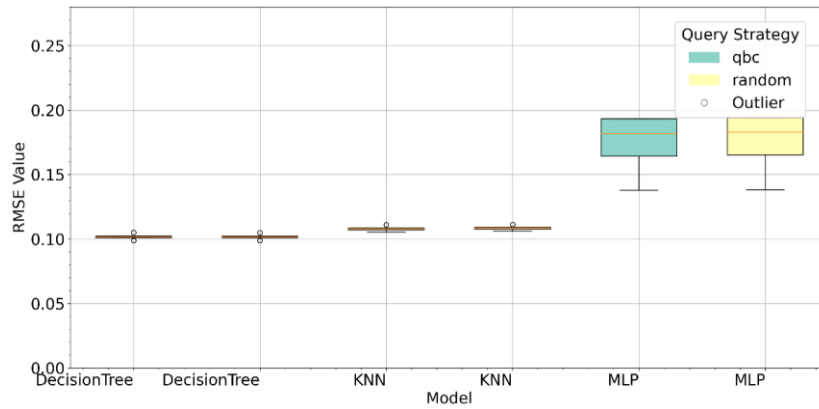


Figure 13: RMSE results for all active learners on the “Voltage” feature.

Figure 14 shows MAE, while Figure 15 shows R^2 score

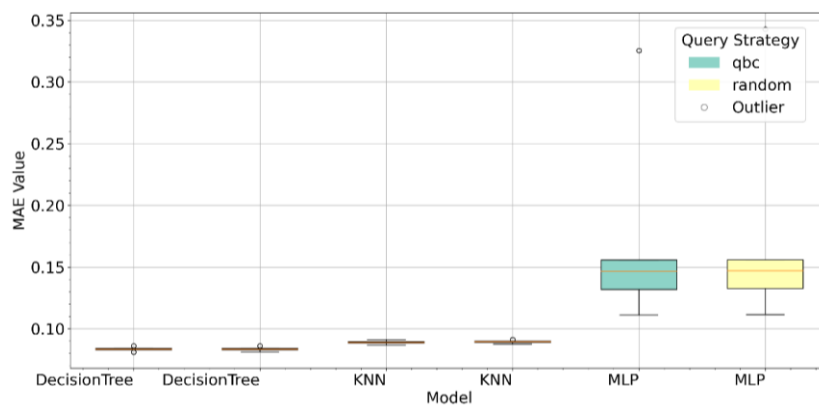
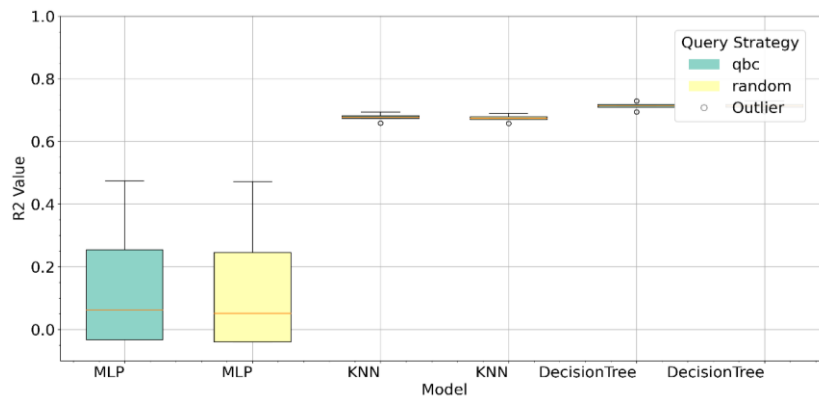


Figure 14: MAE for the “Voltage” feature



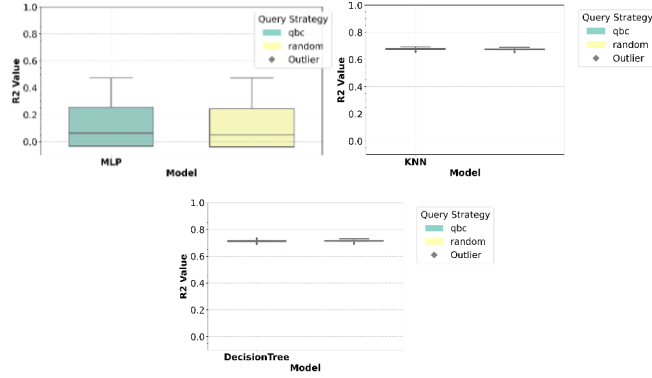


Figure 15: R2 results from all active learners (top) and per model type (bottom) on the Voltage feature from the HVB dataset

Figure 15 shows the clear improvement in R^2 score due to the QBC strategy.

4.5 Material Weight

The results on the Material Weight feature are shown in Table 16.

Table 16: Mean results on Material Weight

Learner (Strategy)		NRMSE		Standard deviation		R2		Standard deviation
KNN (QBC)	•	0.02	•	0.00	•	0.99	•	0.00
KNN (Random)	•	0.02	•	0.00	•	0.99	•	0.00
Decision Tree (Random)	•	0.11	•	0.00	•	0.85	•	0.00
Decision Tree (QBC)	•	0.11	•	0.00	•	0.85	•	0.00
MLP (QBC)	•	0.15	•	0.01	•	0.62	•	0.05
MLP (Random)	•	0.15	•	0.01	•	0.59	•	0.05

The Mann-Whitney U-test results are shown in Table 17.

Table 17: Mann-Whitney U-test ranking results on the “Material Weight” feature from the HVB dataset

Learner (Strategy)		Total Wins		Total Losses		Total Draws		Performance Score
KNN (QBC)	•	5	•	0	•	0	•	1.00
KNN (Random)	•	4	•	1	•	0	•	0.80

Decision Tree (QBC)	•	2	•	2	•	1	•	0.50
Decision Tree (Random)	•	2	•	2	•	1	•	0.50
MLP (QBC)	•	1	•	4	•	0	•	0.20
MLP (Random)	•	0	•	5	•	0	•	0.00

Figure 16, Figure 17, and Figure 18 present the RMSE, MAE, and R^2 distributions, respectively. The improvements caused by the QBC strategy are evident in model types, except for the decision tree learners.

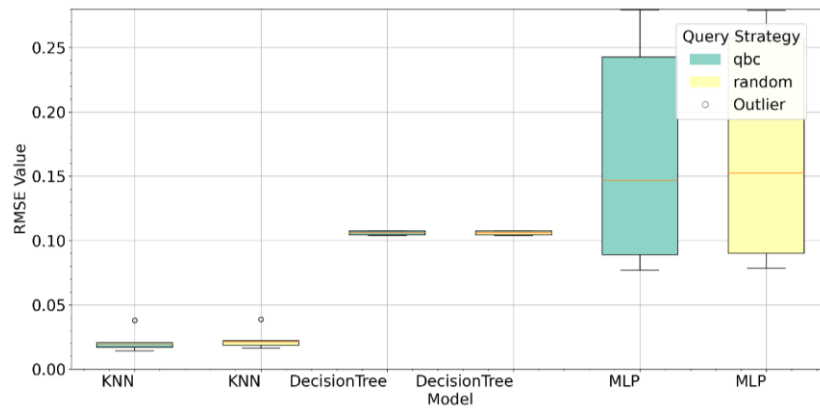


Figure 16: RMSE results for all active learners on the “Material Weight” feature.

Figure 17 and Figure 18 present the MAE, and R^2 distributions, respectively.

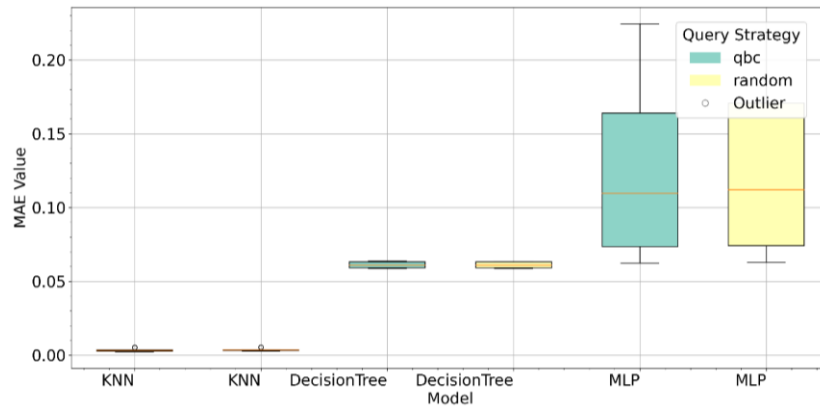


Figure 17: MAE results for all active learners on the “Material Weight” feature.

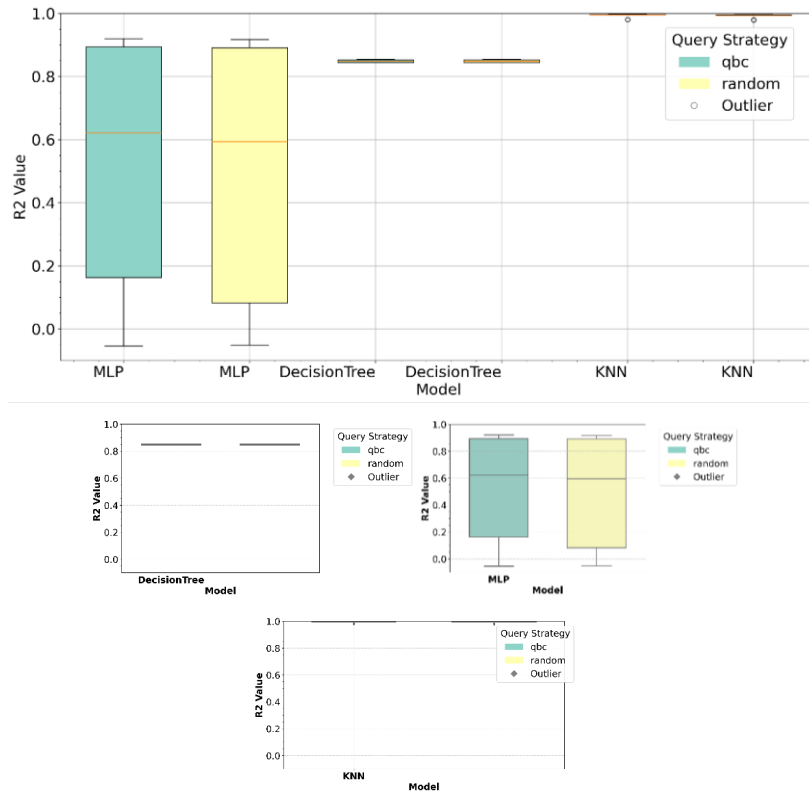


Figure 18: R2 results from all active learners (top) and per model type (bottom) on the Material Weight feature from the HVB dataset

4.6 Material Part Number

[Table 18](#) presents the results of the active learners on the Material Part Number feature.

[Table 18](#): Mean results on Material Part Number

Learner (Strategy)	NRMSE	Standard deviation	R2	Standard deviation
Decision Tree (QBC)	• 0.06	• 0.00	• 0.92	• 0.06
Decision Tree (random)	• 0.06	• 0.00	• 0.92	• 0.06
KNN (QBC)	• 0.09	• 0.00	• 0.81	• 0.09
KNN (Random)	• 0.09	• 0.00	• 0.81	• 0.09

MLP (QBC)	• 0.27	• 0.01	• $\frac{-}{1.66}$	• 0.27
MLP (Random)	• 0.28	• 0.02	• $\frac{-}{2.70}$	• 0.28

Similar to the “Capacity Throughput” feature, the decision tree and k-NN QBC learners outperform their random sampling variants. The MLP learners exhibit low performance on this feature, seen by the negative R^2 scores. Results of the Mann-Whitney U test are shown in Table 19.

Table 19: Mann-Whitney U-test ranking results on the “Material Part Number” feature from the HVB dataset

Model (Strategy)	Total Wins	Total Losses	Total Draws	Performance Score
Decision tree (QBC)	• 5	• 0	• 0	• 1.00
Decision tree (Random)	• 4	• 1	• 0	• 0.80
KNN (QBC)	• 3	• 2	• 0	• 0.60
KNN (Random)	• 2	• 3	• 0	• 0.40
MLP (QBC)	• 0	• 4	• 1	• 0.10
MLP (Random)	• 0	• 4	• 1	• 0.10

The MAE, NRMSE, and R^2 scores are shown in Figure 19, Figure 20, and Figure 21, respectively.

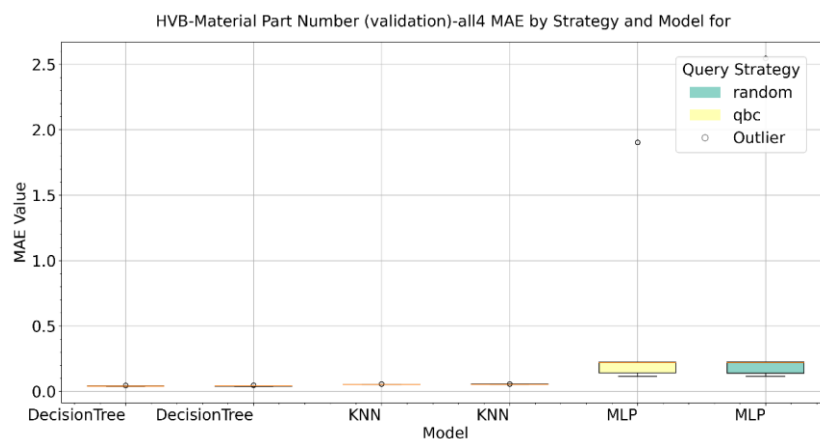


Figure 19: MAE results for all active learners on the “Material Part Number” feature.

The NRMSE, and R^2 scores are shown in Figure 20, and Figure 21, respectively.

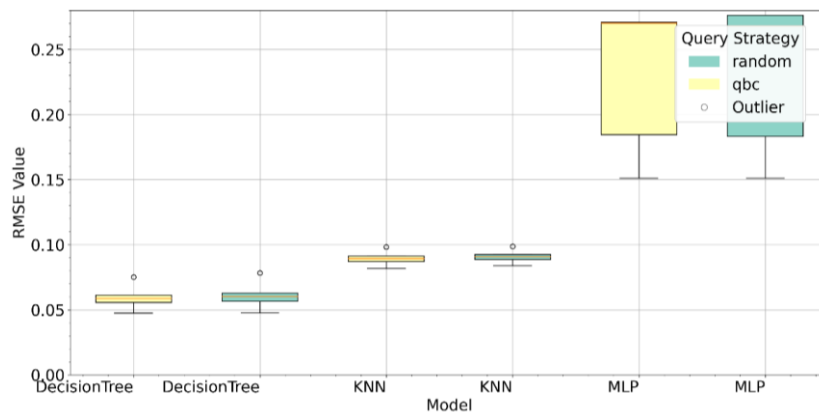
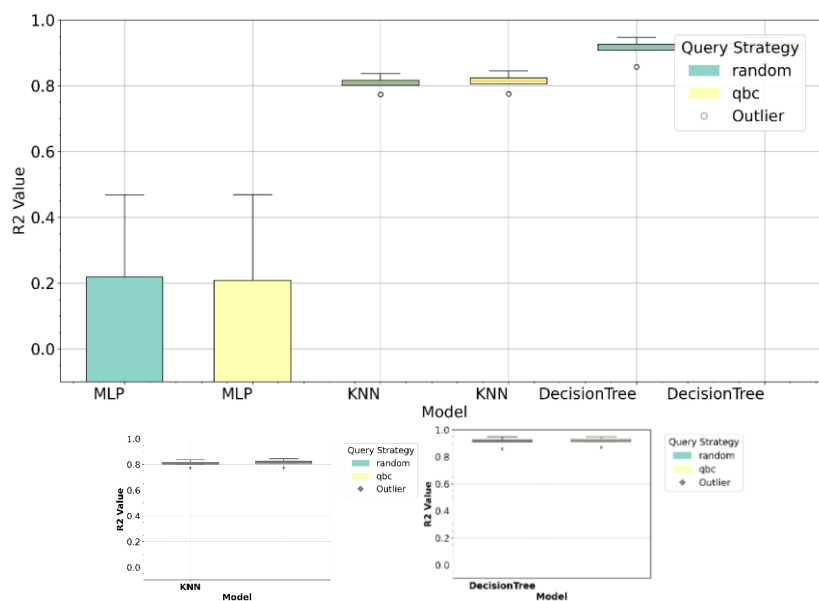


Figure 20: RMSE results for all active learners on the “Material Part Number” feature.



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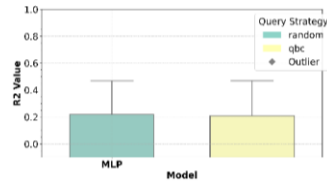


Figure 21: R2 results from all active learners (top) and per model type (bottom) on the Material Part Number feature from the HVB dataset

5. Summary of Results

The extended experiments confirm that the effectiveness of an active learning strategy depends on both the task and data characteristics. Across datasets and features, decision tree-based active learners consistently achieved the highest accuracy and lowest error, demonstrating strong generalisation and robustness to incremental updates. The query-by-committee (QBC) strategy generally improved model performance—particularly for weaker learners such as MLPs—though its advantage diminished in balanced or low-variance datasets where random sampling performed comparably.

Results also show that k-NN learners benefited most from QBC in relative terms, especially on features with smoother or more continuous relationships, such as *Material Weight*. However, the marginal gains were modest compared to those of the decision tree learners. MLP models remained unstable, with QBC mitigating but not eliminating sensitivity to new labels and poor fit on certain features.

On the HVB dataset, QBC outperformed random sampling across most models and features. Decision trees achieved the highest R^2 scores (≥ 0.99) and the lowest NRMSEs on “Minimum Capacity” and “Capacity Throughput” (Table 10 and Table 12), with QBC again slightly outperforming random sampling. KNN also performed well, while MLPs showed limited effectiveness despite optimisation. The rankings are consistent across all metrics for each feature. Each features learner rankings, RMSE, MAE, and R^2 distributions and distributions for R^2 for each model type are included under the relevant subheading.

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On the “Material Part Number” feature (Table 18), decision trees again outperformed other learners, with QBC producing slightly better results than random sampling. MLP models exhibited highly unstable and poor performance, with strongly negative R^2 values, suggesting poor fit.

Across all regression experiments, the decision tree (QBC) active learner consistently demonstrated the best overall performance, outperforming other learners on three of the six features and matching the decision tree (random) learner on two. The k-NN (QBC) learner achieved the highest performance on “Material Weight” and was joint second-best on the “Rings” feature of the abalone dataset.

Overall, the appendix findings reinforce the conclusions of the main study: decision tree and k-NN active learners offer a practical balance between performance, interpretability, and retraining efficiency, while QBC provides a reliable but computationally heavier strategy for improving generalisation. The observed stability across batch sizes further supports the suitability of these models and querying methods for iterative, production-level HITL-ML systems.